

$$6.2) \quad B = \left(\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right)$$

$$\left[\begin{pmatrix} 4 \\ 1 \\ 2 \\ 5 \end{pmatrix} \right]_B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \left[\begin{pmatrix} 2 \\ 2 \\ -2 \\ -2 \end{pmatrix} \right]_B = \begin{pmatrix} 2 \\ -2 \end{pmatrix}$$

$$\Rightarrow [f]_B^B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

Nalezneme vl. čísla $[f]_B^B$

$$\det([f]_B^B - \lambda I_2) = (1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6 = (\lambda+3)(\lambda-2)$$

$$\Rightarrow \lambda_1 = -3, \lambda_2 = 2$$

Nalezneme příslušné vl. vektory

$$\text{Ker}([f]_B^B + 3I_2) = \text{Ker}\left(\begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}\right) = \text{Span}\left\{\begin{pmatrix} -1 \\ 2 \end{pmatrix}\right\}$$

$$\Rightarrow [\vec{v}_1]_B = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \Rightarrow \vec{v}_1 = \begin{pmatrix} 0 \\ -1 \\ 2 \\ 3 \end{pmatrix}$$

$$\text{Ker}([f]_B^B - 2I_2) = \text{Ker}\left(\begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix}\right) = \text{Span}\left\{\begin{pmatrix} 2 \\ 1 \end{pmatrix}\right\}$$

$$\Rightarrow [\vec{v}_2]_B = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \Rightarrow \vec{v}_2 = \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix}$$

$C = (\vec{v}_1, \vec{v}_2)$ je báze prostoru V

$$\left[\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right]_C = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\left[f^n \left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right) \right]_C = [f^n]_C \left[\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right]_C = \begin{pmatrix} (-3)^n & 0 \\ 0 & 2^n \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} (-3)^n \\ 2^n \end{pmatrix}$$

$$\Rightarrow f^n \left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right) = (-3)^n \begin{pmatrix} 0 \\ -1 \\ 2 \\ 3 \end{pmatrix} + 2^n \begin{pmatrix} 5 \\ 2 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \cdot 2^n \\ -(-3)^n + 2 \cdot 2^n \\ 2 \cdot (-3)^n + 2^n \\ 3 \cdot (-3)^n + 4 \cdot 2^n \end{pmatrix}$$